

Official Scraped Data:

Major_Requirements

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Computer Science, B.S.

The undergraduate Computer Science curriculum emphasizes the principles of computing that underpin our modern world, and provides a strong foundation to prepare students for the broad spectrum of careers in computing including within industry, academia, government, and non-profit organizations.

The Computer Science major is highly customizable, enabling students to explore the diverse modern computing landscape. The lower-division core requirements provide a solid foundation in programming methodologies, computer architecture, middle-level infrastructure, data structures, algorithms, and mathematical fundamentals of computing. The upper-division requirements include flexible core electives that expose the students to the breadth of areas within computer science, along with additional electives that allow students to pursue selected areas in greater depth. To ensure that the students build strong skills in design and implementation, the curriculum requires a minimum of two project-based courses.

For suggested upper-division courses within the different areas of Computer Science, refer to the 'Upper-Division Areas of Study' section.

The department also offers a joint undergraduate degree in Computer Science and Engineering, in conjunction with The Henry Samueli School of Engineering; information is available in the Interdisciplinary Studies

section of the Catalogue.

To learn more about the department, including research areas and latest news, visit the Computer Science department website .

Freshman Applicants: See the Undergraduate Admissions section .

Transfer Applicants:

Transfer applicants who satisfactorily complete course prerequisites will be given preference for admission. All applicants must complete the following required courses: one year of approved calculus, one year of object-oriented programming (python, java, C++), additional courses as specified by the major, and completion of lower-division writing. Students are encouraged to complete as many of the lower-division degree requirements as possible prior to transfer. Visit the UCI Office of Admissions website for information on transfer requirements for our major.

Requirements for the B.S. in Computer Science

All students must meet the University Requirements .

Major Requirements

Lower-division

A. Select one of the following series:

I&C SCI 31

32

33

Introduction to Programming and Programming with Software Libraries and Intermediate Programming

or

I&C SCI H32

33

Python Programming and Libraries (Accelerated) and Intermediate Programming

B. Complete:

I&C SCI 45C

Programming in C/C++ as a Second Language

I&C SCI 46

Data Structure Implementation and Analysis

I&C SCI 51

Introductory Computer Organization

I&C SCI 53

Principles in System Design

SWE 43

Introduction to Software Engineering

MATH 2A

2B

Single-Variable Calculus I and Single-Variable Calculus II

I&C SCI 6B

Boolean Logic and Discrete Structures

I&C SCI 6D

Discrete Mathematics for Computer Science

I&C SCI 6N

Computational Linear Algebra

or MATH 3A

Introduction to Linear Algebra

STATS 67

Introduction to Probability and Statistics for Computer Science

C. Two courses approved for General Education category II except those offered by the School of Engineering, the School of Information and Computer Sciences, the Department of Economics, or the Department of Mathematics. University Studies courses can be used with the approval of the Computer Sciences Vice Chair for Undergraduate Studies.

Upper-division (17 courses)

Complete sections A-D. Courses used in one section may not apply to another, except CS 145 which may apply to sections B and D.

A. Core (2 courses)

COMPSCI 161

Design and Analysis of Algorithms

I&C SCI 139W

Critical Writing on Information Technology

B. Flexible Core Requirement (4 courses)

Complete one course from at least four distinct focus areas (FA) listed

below.

FA1: Vision and Graphics

COMPSCI 112

Computer Graphics

COMPSCI 116

Computational Photography and Vision

FA2: Information Management

COMPSCI 121

Information Retrieval

COMPSCI 122A

Introduction to Data Management

FA3: Networks and Cybersecurity

COMPSCI 130

Introduction to Security and Privacy

COMPSCI 132

Computer Networks

COMPSCI 134

Computer and Network Security

FA4: Systems and Software

COMPSCI 141

Concepts in Programming Languages

COMPSCI 142A

Compilers and Interpreters

COMPSCI 143A

Principles of Operating Systems

FA5: Architecture and Embedded Systems

COMPSCI 145

Embedded Software

COMPSCI 151

Digital Logic Design

COMPSCI 152

Computer Architecture

FA6: Algorithms

COMPSCI 162

Formal Languages and Automata

COMPSCI 163

Graph Algorithms

COMPSCI 164

Computational Geometry

COMPSCI 167

Introduction to Applied Cryptography

FA7: Artificial Intelligence and ML

COMPSCI 171

Introduction to Artificial Intelligence

COMPSCI 178

Machine Learning and Data-Mining

COMPSCI 184A

Artificial Intelligence in Biology and Medicine

C. Upper-division electives (9 courses):

Complete nine courses from the list below.

COMPSCI 103 -189

SWE 113

Requirements Analysis and Engineering

SWE 115

Software Testing, Analysis, and Quality Assurance

SWE 117

Project in Software System Design

SWE 121

Software Design: Applications

SWE 122

Software Design: Structure and Implementation

SWE 124

Internet Applications Engineering

IN4MATX 131

Human Computer Interaction

IN4MATX 133

User Interaction Software

IN4MATX 134

Advanced Topics in User Interface Software

D. Project Courses (2 courses):

Complete two courses from the list below.

COMPSCI 114

Projects in Advanced 3D Computer Graphics

COMPSCI 117

Project in Computer Vision

COMPSCI 118

Introduction to Virtual Reality

COMPSCI 122B

Project in Databases and Web Applications

COMPSCI 122C

Principles of Data Management

COMPSCI 122D

Beyond SQL Data Management

COMPSCI 125

Next Generation Search Systems

COMPSCI 133

Advanced Computer Networks

COMPSCI 142B

Project in Compilers and Interpreters

COMPSCI 143B

Project in Operating System Organization

COMPSCI 145

Embedded Software

COMPSCI 147

Internet of Things (IoT) Software and Systems

COMPSCI 154

Computer Design Laboratory

COMPSCI 165

Project in Algorithms and Data Structures

COMPSCI 175

Project in Artificial Intelligence

COMPSCI 180A

Project in Computer Science

COMPSCI 180B

Project in Computer Science

COMPSCI 189

Project in Bioinformatics

SWE 117

Project in Software System Design

IN4MATX 134

Advanced Topics in User Interface Software

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Students must complete and pass both COMPSCI 180A and COMPSCI 180B for course credit.

Computer Science is a broad field of study. The Computer Science degree curriculum encourages students to tailor their upper-division course work toward their specific areas of interest, or alternatively allows students to explore any number of areas. Areas of interest include:

Algorithms , including theory of computation, graph and network algorithms, probabilistic algorithms, computational geometry, cryptography, and quantum computation.

Related courses: COMPSCI 162

COMPSCI 169

Complementary courses: COMPSCI 177 , COMPSCI 179

Architecture and Embedded Systems , including hardware, digital logic design, computer architecture, embedded computing systems, and multitasking operating systems.

Related courses: COMPSCI 145

COMPSCI 154

Complementary courses: COMPSCI 131

Artificial Intelligence and Machine Learning , including human knowledge representation, building automated reasoning systems, developing intelligent search techniques and algorithms that adapt and learn from data, with applications to bioinformatics, and beyond.

Related courses: COMPSCI 171

COMPSCI 179

Complementary courses: COMPSCI 116 , COMPSCI 121

COMPSCI 125 , COMPSCI 169 , COMPSCI 183 , COMPSCI 184A

COMPSCI 189

Information Management , including information retrieval, database management, storage management, data-intensive computing, and next generation search systems.

Related courses: COMPSCI 121

COMPSCI 125 , COMPSCI 178

Complementary courses: COMPSCI 177 , COMPSCI 179

Networks and Cybersecurity , including Internet architecture, Internet applications, distributed systems, network security, computer security, and privacy.

Related courses: COMPSCI 130

COMPSCI 137

Complementary courses: COMPSCI 143A

COMPSCI 143B , COMPSCI 167

Systems and Software , including programming languages, compilers and interpreters, operating systems, and parallel and distributed computing.

Related courses: COMPSCI 131 , COMPSCI 141

COMPSCI 143B

Complementary courses: COMPSCI 103 , COMPSCI 146

COMPSCI 147

Visual Computing , including computer graphics, 3D graphics, image processing, image recognition, computational photography, computer vision, and virtual reality.

Related courses: COMPSCI 111

COMPSCI 114 , COMPSCI 116

COMPSCI 118

Complementary courses: COMPSCI 115

Freshman

Fall

Winter

Spring

I&C SCI 31 I&C SCI 32 I&C SCI 33

MATH 2A MATH 2B SWE 43

WRITING 40 WRITING 50 I&C SCI 6B

General Education III WRITING 60

Sophomore

Fall

Winter

Spring

I&C SCI 45C I&C SCI 46 Flex Core Requirement

I&C SCI 51 I&C SCI 6N or MATH 3A Flex Core Requirement

I&C SCI 6D General Education III STATS 67

General Education IV/VIII General Education III/VII

Junior

Fall

Winter

Spring

I&C SCI 53 Flex Core Requirement Computer Science Elective

COMPSCI 161 Flex Core Requirement Computer Science Elective

Science Elective I&C SCI 139W Computer Science Elective

Science Elective General Education VI

Senior

Fall

Winter

Spring

Computer Science Elective Computer Science Elective Computer Science Elective

Computer Science Elective Computer Science Elective Computer Science Elective

CS Project CS Project General Education IV

General Education IV

NOTES :

1. Students are advised that this sample program lists the minimum requirements; it is possible that students may have to take additional courses to prepare for required courses.
2. The lower-division writing requirement must be completed by the end of the seventh quarter at UCI.
3. This is only a sample plan. Course offerings may be moved due to unforeseen circumstances. It is strongly recommended that students meet with an academic advisor to create an academic plan tailored to meet their specific areas of interest.

Computer Science, M.S. Computer Science, Ph.D. Master of Computer Scienc